Optimal Entropy Cost of Uniform Causal Approximation

A sharp accuracy law for vanishing binary activity

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Abstract

For a locally finite binary controlled tree with attainable envelope $p_t = 1/[t(\ln t)^2]$, we prove that the minimum Shannon entropy of a common causal seed approximating every infinite policy transcript within total variation ε is $\log_2(1/\varepsilon) + O(1)$. The upper bound is attained up to an additive constant by finite seeds. The converse applies to all countably supported seeds and follows already from the last-success time of one maximizing policy. Thus the upper bound for temporal truncation is optimal in its leading coefficient, even among arbitrary common causal decoders. This is a mathematical proof, not a Lean-certified extension; publication priority is not asserted.

1 Model and statement

Time is indexed by integers $t \geq 3$. At every finite action–output history h, a finite nonempty action set is available. The output is a bit, with conditional success probability $q_t(h, a)$. Assume

$$0 \leq q_t(h, a) \leq p_t := \frac{1}{t(\ln t)^2}, \qquad \max_a q_t(h, a) = p_t \quad \text{for every } t, h. \tag{1}$$

Policies are deterministic and adaptive. A seed S is a countably supported random variable, independent of the policy choice, whose value specifies a total deterministic response strategy on the tree. The decoder uses only the seed, the current history and the current action. No extra random input is permitted. Kernels and decoder descriptions are not charged to $H(S)$.

Write P_π for the prescribed infinite transcript law and $\widehat{P}_{\pi,S}$ for the law generated by the seed under policy π. Total variation is normalized as $\mathrm{TV}(P, Q) = \sup_A |P(A) - Q(A)|$; on a countable space it is $\frac{1}{2}\sum_x |P_x - Q_x|$. Define

$$C_{\mathrm{app}}(\varepsilon) = \inf\left\{ H(S) : \sup_\pi \mathrm{TV}(P_\pi, \widehat{P}_{\pi,S}) \leq \varepsilon \right\}.$$

All entropies are in bits, all unmarked logarithms are natural, and $0 \log_2 0 = 0$. The error constraint concerns entire infinite transcripts, not just a fixed horizon.

Theorem 1. *Under* (1)*, there are finite constants* K_-, K_+ *and* $\varepsilon_0 > 0$ *such that, for* $0 < \varepsilon \leq \varepsilon_0$*,*

$$\boxed{\log_2(1/\varepsilon) - K_- \leq C_{\mathrm{app}}(\varepsilon) \leq \log_2(1/\varepsilon) + K_+.}$$

The upper bound uses finite seeds. The lower bound holds for arbitrary countably supported seeds, even if correctness is required only for one maximizing policy. The constants may be chosen independently of the number of actions and of the history dependence of the kernels.

2 An entropy lower bound under total-variation perturbations

Lemma 1. *Let P, Q be probability distributions on a common countable alphabet with $\mathrm{TV}(P,Q) \le \varepsilon$. For every $M > 0$,*

$$H(Q) \ge \mathbb{E}_P \min\{\log_2(1/P(X)), M\} - \varepsilon M - 1. \tag{2}$$

Zero-mass points contribute zero to the expectation.

Proof. Put $r_x = \min(P_x, Q_x)$, $s_x = Q_x - r_x$ and $\delta = \sum_x s_x = \mathrm{TV}(P,Q)$. Introduce a binary label distinguishing the subprobabilities r and s. Its marginal entropy is $h_2(\delta) \le 1$. The entropy chain rule gives

$$H(Q) + h_2(\delta) \ge \sum_x r_x \log_2(1/r_x) + \sum_x s_x \log_2(1/s_x).$$

This inequality is valid for extended entropies, for example by the usual finite-partition approximation. If $H(Q) = \infty$, the claim is immediate. Otherwise the displayed inequality also ensures that the right side is finite. Since each $s_x \in [0,1]$, its entropy contribution is nonnegative. For $P_x > 0$, $r_x \le P_x$ gives

$$r_x \log_2(1/r_x) \ge r_x \log_2(1/P_x) \ge r_x \min\{\log_2(1/P_x), M\}.$$

Finally, $\sum_x (P_x - r_x) = \delta$, so replacing r by P in the last bounded sum increases it by at most $\delta M \le \varepsilon M$. □

3 The last-event distribution

Choose a deterministic policy $\pi^\star$ attaining p_t at every history. Its outputs are independent $B_t \sim \mathrm{Bernoulli}(p_t)$. Since $\sum_{t \ge 3} p_t < \infty$, the last-success time

$$J = \max\big(\{t \ge 3 : B_t = 1\} \cup \{0\}\big)$$

is finite almost surely. Let

$$A = \prod_{t=3}^{\infty}(1 - p_t) > 0, \qquad r_0 = A, \qquad r_j = p_j a_j, \quad a_j = \prod_{t>j}(1 - p_t) \quad (j \ge 3).$$

These are the masses of J. Positivity of A follows from summability of p_t and $p_t \le 1/2$.

Lemma 2. *As $N \to \infty$,*

$$\sum_{j=3}^{N} r_j \log_2(1/r_j) = \log_2 \ln N + O(1). \tag{3}$$

Moreover, for $3 \le j \le N$,

$$r_j \ge \frac{A}{N(\ln N)^2}. \tag{4}$$

Proof. The integral test and the union bound give

$$0 \le 1 - a_j \le \sum_{t>j} p_t \le \frac{1}{\ln j}.$$

For all sufficiently large j, $a_j \geq 1/2$, and hence $-\ln a_j \leq 2(1-a_j)$. With $f(u) = u\log_2(1/u)$,

$$f(p_j a_j) - f(p_j) = -p_j(1-a_j)\log_2(1/p_j) + p_j a_j \log_2(1/a_j).$$

Because $\log(1/p_j) = \ln j + 2\ln\ln j$, the absolute value of this difference is $O(1/[j(\ln j)^2])$, a summable sequence. On the other hand,

$$f(p_j) = \frac{1}{\ln 2}\left(\frac{1}{j\ln j} + \frac{2\ln\ln j}{j(\ln j)^2}\right).$$

The second series converges, while the first has partial sums $\ln\ln N + O(1)$. This proves (3). Finally $a_j \geq A$, and $j(\ln j)^2$ is increasing for $j \geq 3$, giving (4). □

4 Converse for arbitrary seeds

Let S satisfy the approximation requirement. Under $\pi^\star$, apply to its generated transcript the measurable map returning the last-success time, or $+\infty$ if infinitely many successes occur. Denote the resulting random variable by $\widehat{J}$ and its law by Q. Extend the law P of J by assigning mass zero to $+\infty$. Data processing gives

$$\mathrm{TV}(P,Q) \leq \varepsilon, \qquad H(S) \geq H(\widehat{J}).$$

For $N = \lceil e^{1/\varepsilon} \rceil$, set

$$M_N = \log_2 \frac{N(\ln N)^2}{A}.$$

By (4), all atoms r_j with $3 \leq j \leq N$ have information at most M_N. Therefore Lemmas 1 and 2 imply

$$H(S) \geq \sum_{j=3}^{N} r_j \log_2(1/r_j) - \varepsilon M_N - 1 \geq \log_2\ln N - O(1) - \varepsilon M_N.$$

Here $\ln N = 1/\varepsilon + o(1)$ and

$$\varepsilon M_N = \frac{1}{\ln 2}\left(\varepsilon\ln N + 2\varepsilon\ln\ln N - \varepsilon\ln A\right) = O(1).$$

Consequently $H(S) \geq \log_2(1/\varepsilon) - O(1)$. Taking the infimum establishes the lower bound in Theorem 1.

5 Achievability by one finite causal seed

For completeness we give the construction. Independently at each time, sample a coordinate with law

$$\mathbb{P}(Z_t = 0) = 1 - p_t, \qquad \mathbb{P}(Z_t = k) = p_t 2^{-k} \quad (k \geq 1).$$

Choose binary digits for every history and action so that $q_t(h,a)/p_t = \sum_{k\geq 1} b_{t,k}(h,a)2^{-k}$; the value one uses the all-ones expansion. Return zero when $Z_t = 0$, and otherwise return $b_{t,Z_t}(h,a)$. The coordinate Z_t is independent of every earlier history, so this one deterministic response table realizes exactly every adaptive policy. Its coordinate entropy is

$$H(Z_t) = h_2(p_t) + 2p_t.$$

For an integer N, use this table through time N and return zero at all later nodes. Only finitely many response tables exist through depth N, so pushing $(Z_3, \dots, Z_N)$ onto those tables produces a finite seed S_N with

$$H(S_N) \le \sum_{t=3}^{N} [h_2(p_t) + 2p_t].$$

Couple this approximation to the full construction by the same coordinates. On $\{Z_t = 0 \text{ for all } t > N\}$, the entire response tables agree. Thus one coupling works simultaneously for all policies, and

$$\sup_{\pi} \mathrm{TV}(P_\pi, \widehat{P}_{\pi,S_N}) \le \sum_{t>N} p_t \le \int_N^\infty \frac{dx}{x(\ln x)^2} = \frac{1}{\ln N}.$$

Since $h_2(p) = p \log_2(1/p) + O(p)$ as $p \downarrow 0$, the estimates above also give

$$\sum_{t=3}^{N} [h_2(p_t) + 2p_t] = \log_2 \ln N + O(1).$$

Taking $N = \lceil e^{1/\varepsilon} \rceil$ completes the upper bound.

6 Scope and interpretation

The result is about entropy, not seed support size, running time, or the description length of the kernels. Its upper construction uses a horizon exponential in $1/\varepsilon$ and makes no computational-efficiency claim. For each fixed error tolerance the seed is finite, although every exact infinite-horizon countable seed has infinite entropy in this model.

The converse uses only one observable of one policy. The same leading entropy cost therefore governs approximation of that last-event law and uniform causal approximation of the entire policy family. This does not make the last-event time a sufficient seed for the full process.

Minimizing Shannon entropy over a total-variation ball is an established smooth-entropy problem; variable-length resolvability provides relevant precedents [1]. The present note supplies an explicit converse and matching construction for the stated sparse adaptive model. Neither historical priority nor Lean verification of this extension is claimed. The repository's current formal declarations must be checked separately from the mathematical proof given here.

References

[1] *Variable-Length Resolvability for General Sources and Channels*. Entropy **25**(10), 1466 (2023). `https://doi.org/10.3390/e25101466`.

[2] J. Mikulík. *One Seed for Every Adaptive Policy*. `https://github.com/jendamikulik/finite-Shannon-theorem`.

[3] J. Mikulík. *Wholeness as Lossless Causal Compression*. Mathematical draft, 20 October 2026.